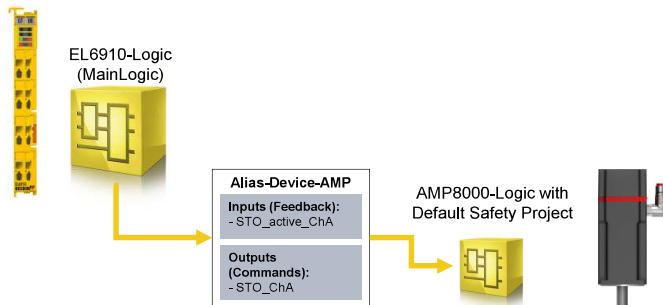


Initial project for ELM, AMI, AMP

Initial Projects

BECKHOFF

- Initial Project = STO-Project of the device in its delivery condition
- Available on the homepage
 - AMP8000
 - AMI8000
 - ELM72xx-9016 and ELM72xx-9018
- Advantage:
 - Only one bit required to command STO



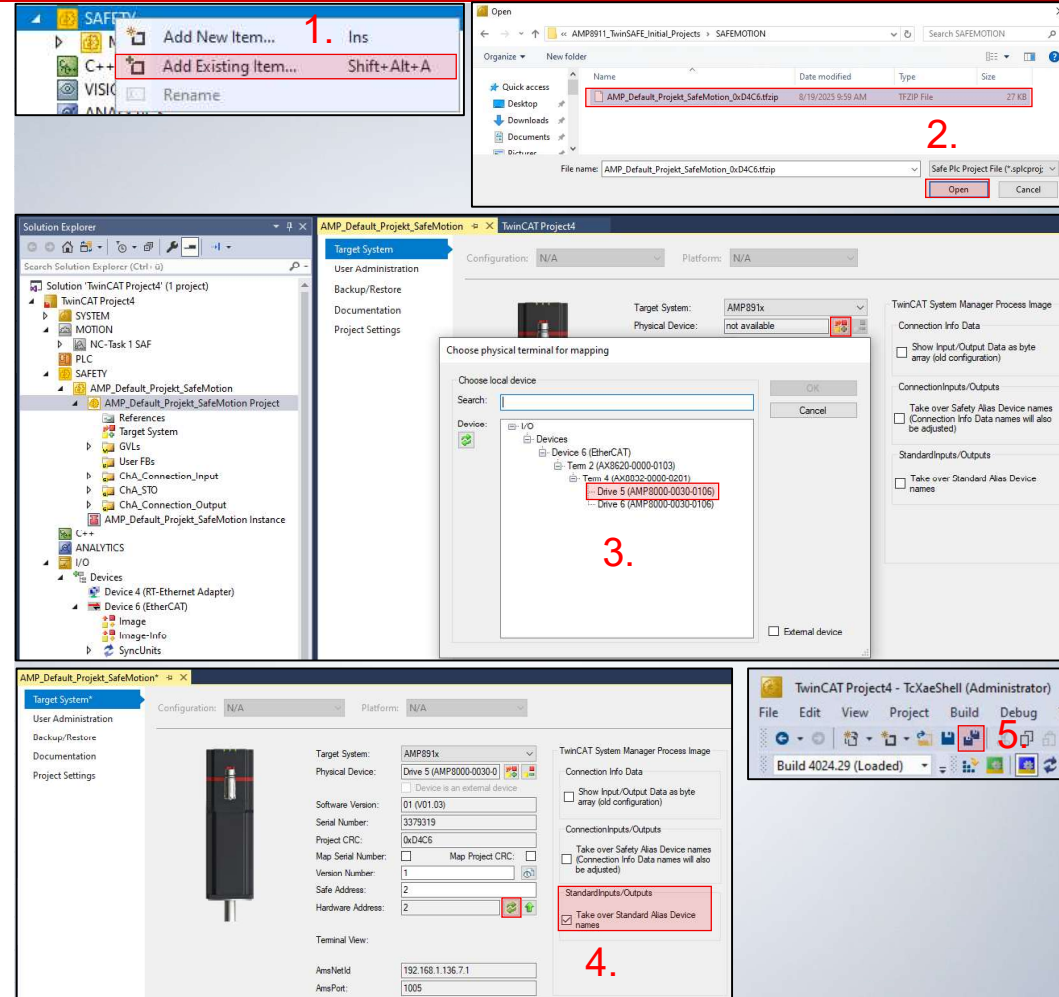
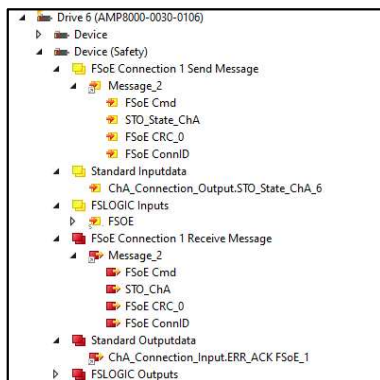
- Default system manager configuration (no adjustment of the safety project image of the drive hardware)
- Simple connection setup via import of alias device

The image contains two screenshots. The top screenshot shows a search results page for 'Initial Project' on the Beckhoff website, displaying 118 results. The bottom screenshot shows the TwinCAT software interface. On the left, a tree view shows the 'Alias-Device-AMP' configuration. On the right, a 'Select from I/O tree' dialog box is open, showing a list of I/O devices. The 'Alias-Device-AMP' is selected, and the 'Import Alias-Device(s) from I/O-configuration' option is highlighted.

Initial Projects | Handling

BECKHOFF

- Download of the initial project from the website
- Inserting the initial project into the Safety node
- Selecting the target system with the hardware
- Update the address and save the project
 - Saving changes the default process image to the initial project process image

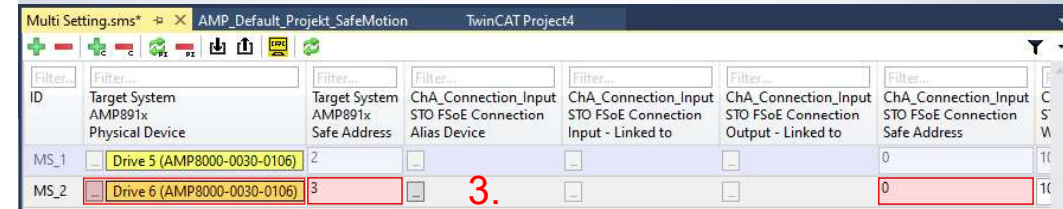
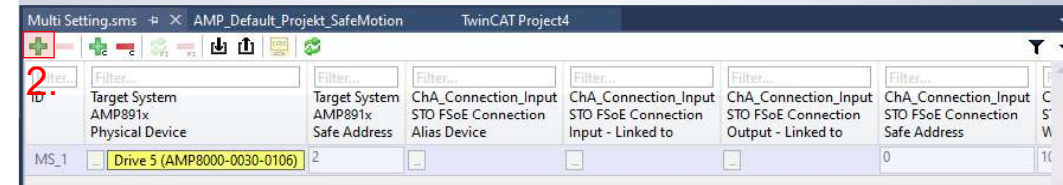
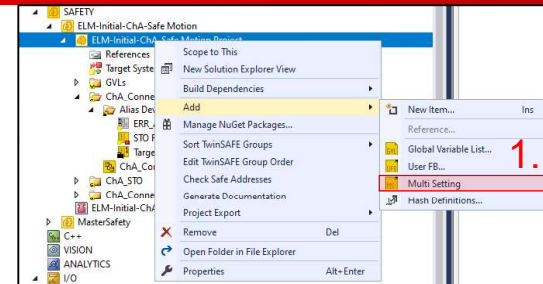


Initial Projects with Multisettings | Handling

BECKHOFF

For multiple axes, use the multi-settings in combination with the initial project.

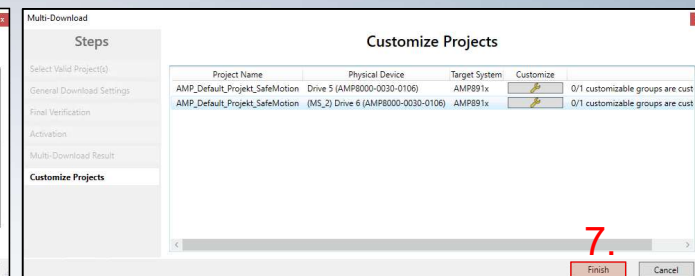
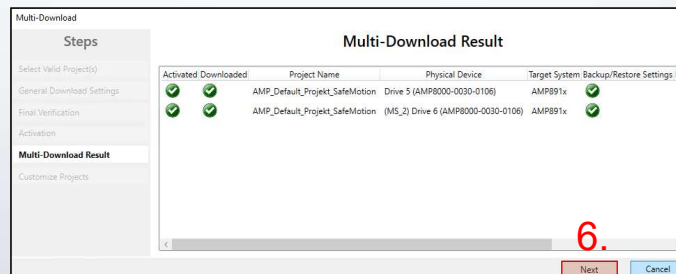
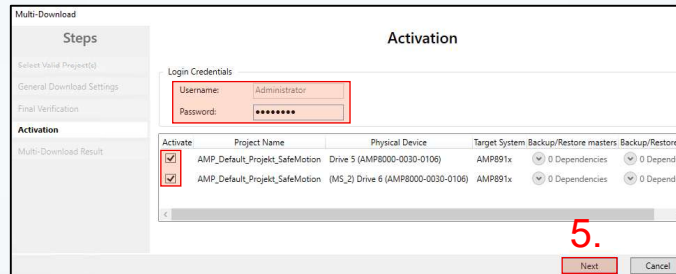
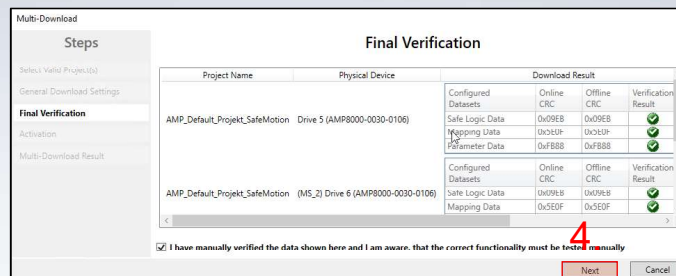
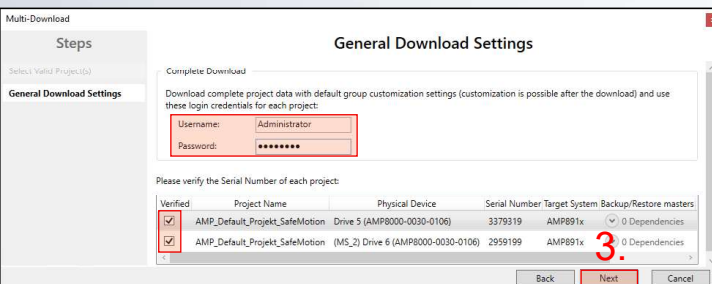
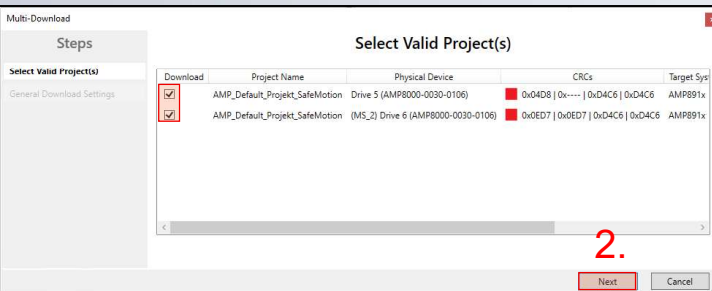
- Adding the multisettings list
- Adding new lines - depending on the number of devices that are to receive the initial project download
- Connecting the devices and entering the safety addresses
→ It is best to change the addresses in advance.
- Update the multisettings list and store the project
 - This changes the process image of the connected devices in the multisettings list



Initial Projects with Multisettings | Handling

BECKHOFF

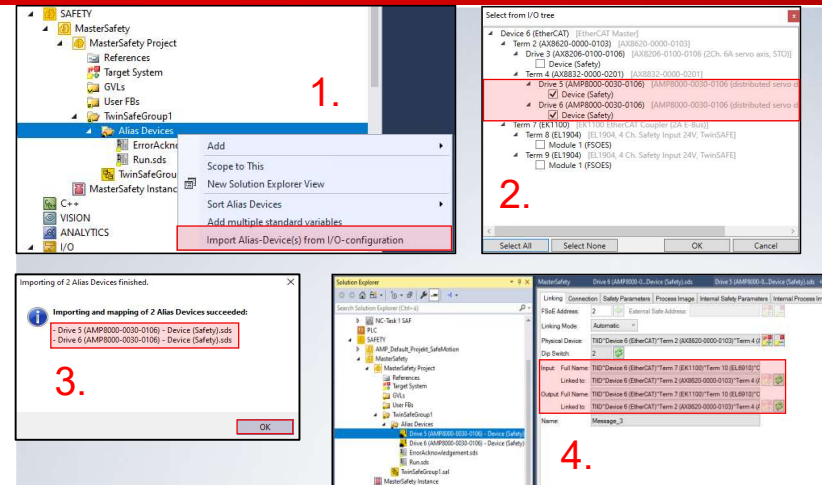
Download the initial project to the selected devices



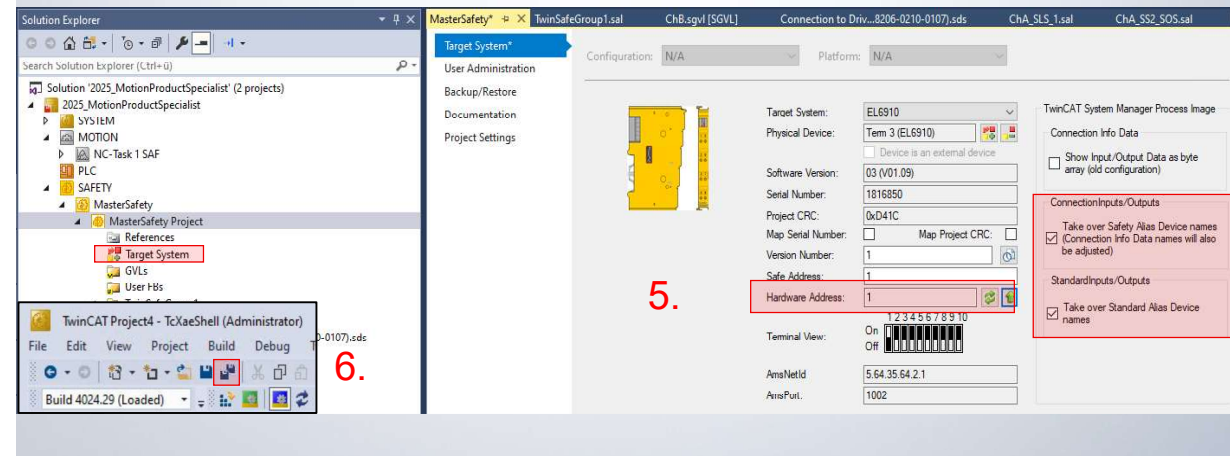
Master safety project in combination with initial project | Handling

BECKHOFF

- Importing the AMP8000 alias device from the IO configuration in the master safety project
- The connections to the AMP8000 SafetyDevice are established automatically



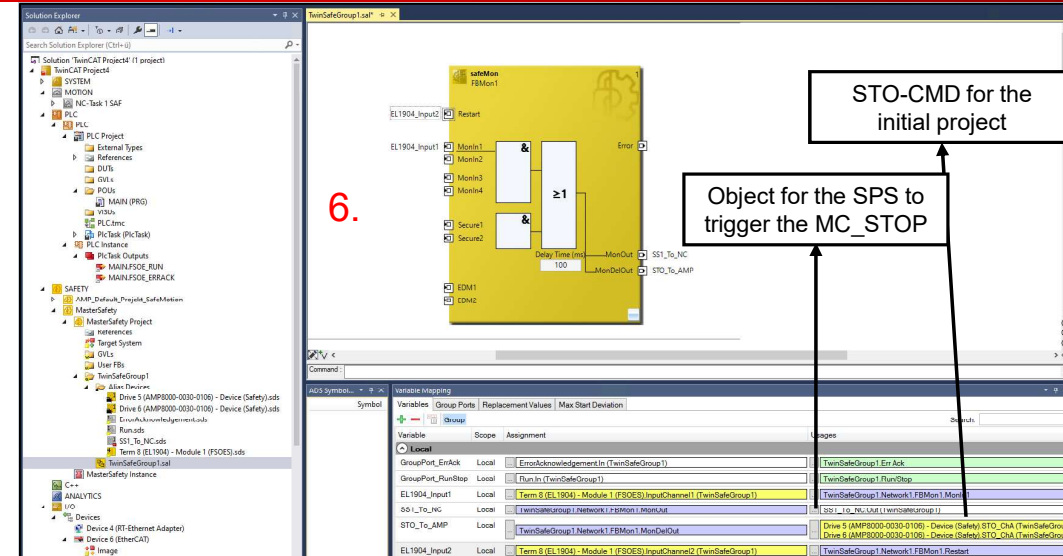
- Update the target system in the master safety project
 - Transfer the hardware address
 - Activate the checkboxes
 - Save the project



Master safety project in combination with initial project | Handling

BECKHOFF

- Configure logic in the MasterSafetyProjekt - TwinSafeGroup and link the corresponding objects
 - STO-Object of the initial project
 - If necessary → SS1-Object for the SPS



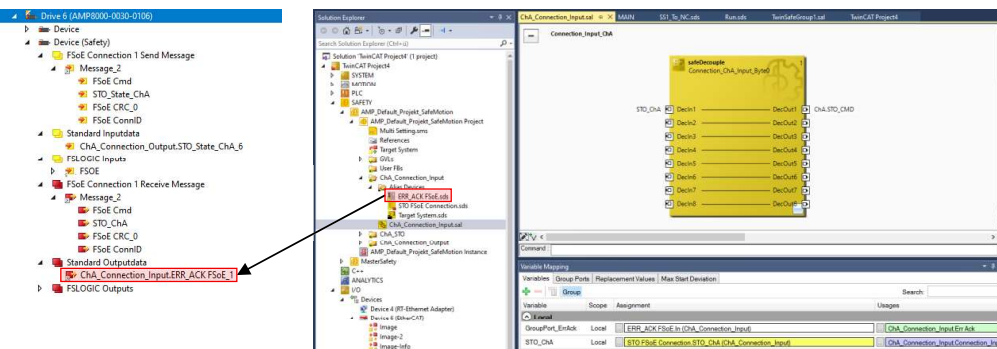
Master safety project in combination with initial project | Handling

BECKHOFF

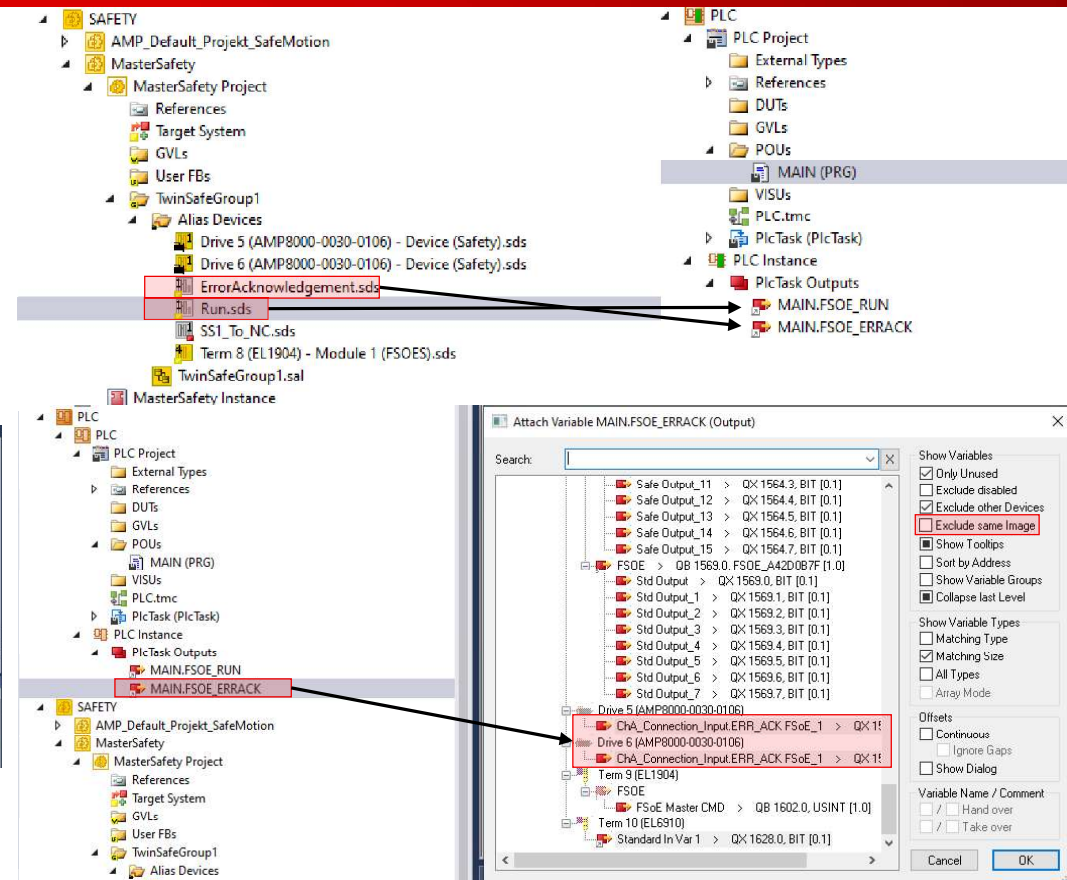
- Linking the FSoE_Run and FSoE_ErrACK variables for switching on and acknowledging the safety connections

Important:

The initial projects have an ErrACK_FSoE variable in their standard initial process image, which must be connected in order to acknowledge the initial project and the connection



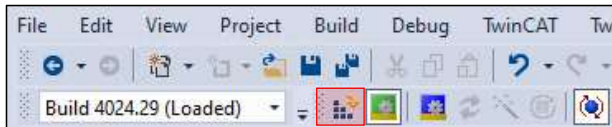
Manual linking of the object necessary → The same variable can be used as in the Master safety project



Master safety project in combination with initial project | Handling

BECKHOFF

- Download the master safety project
- Activation of the configuration



- Set the PLC variables:
FSOE_Run = TRUE
FSOE_ErrACK = FALSE – TRUE – FALSE
- Activate STO via input 1 and acknowledge via input 2
- Check the object:

– AX8000 and AMP8000:

Index	Name	Flags	Value
205F.0	Safety info	RO	> 1 <
3000.0	Ch A AxisMain parameters	RO	> 15 <
3001.0	Ch A AxisMain actual values	RO	> 8 <
3001.03	Missing output cycle counter incre...	RO P	0x00000000 (0)
3001.04	Output stage safety state	RO P	ready_state (1)
3001.05	Dyninput cycle counter	RO P	0x00 (0)
3001.06	Missing dynoutput cycle counter incre...	RO P	0x00000000 (0)
3001.07	Actual motor brake state	RO P	Motor_brake_locked (0)
3001.08	Current error reaction	RO P	Reaction_None (0)

ELM7xxx and AMI8000:

Index	Name	Flags	Value
9010.0	DRV Info data Ch.1		> 39 <
9010.13	Supported drive modes	RO	0x00000780 (1920)
9010.14	Velocity encoder resolution	RO	0x00041893 (268435)
9010.15	Position encoder resolution increments	RO	0x00100000 (1048576)
9010.16	Position encoder resolution revolutions	RO	0x00000001 (1)
9010.17	Cogging compensation supported	RO	TRUE
9010.27	Output stage safety state	RO P	ready_state (1)

